

ECEN-635: Homework 3

Due on September 19, 2025 at 11:59 pm

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Problem 1

The field $\mathcal{E}(t) = 5 \sin(20 \times 10^9 t + kz) \hat{\mathbf{y}}$ (V/m) describes a wave propagating in a medium which has the permittivity of free space with $\epsilon_r = 3$. Answer the following questions concerning this wave. Be sure to include dimensions in your answers:

- (a) What is the frequency f of the wave?
- (b) What is the wave amplitude?
- (c) What is the propagation direction?
- (d) What is the phase velocity of the wave v_p ?
- (e) What is the wavelength λ of the wave?
- (f) What is the intrinsic impedance of the wave η ?
- (g) What is the wave number (propagation constant) k ?

Solution

Problem 2

In lossy insulating material a 2.45 GHz plane wave is propagating in a casserole in the z - direction and polarized in the y -direction, with an amplitude of 20 V/m, a conductivity $\sigma = 2.0$ S/m, and a dielectric constant $\epsilon_r = 4$. Find the loss tangent. Hint: this is a trivial question but the loss tangent is of primary importance when ordering a material.

Solution

Problem 3

The first step in analytically solving for the fields in any situation in electromagnetics is to obtain the vector Helmholtz equation from Maxwell's equations. In class, starting with the Maxwell's equations the electric field vector Helmholtz equation was derived. Starting with Maxwell's equations derive the vector Helmholtz equation for the magnetic field. Write down and number Maxwell's equations and at each step in the derivation tell which equation you are going to use for the next step. If you use a vector identity, before using it write down the identity from those in the back cover of the book. Be sure to use the proper vector symbol and a line above the letter to indicate each vector.

Solution

Problem 4

In each case below (i) show mathematically that the field given is a straight line, a circle or an ellipse (ii) find an expression for the angle between the field vector and the x-axis. (iii) In each case describe the type of polarization. Hint: Remember the polarization is found by finding the phase of one of field components with respect to the other. A trick to find the difference in phase is to add or subtract a phase to both field components. By doing this you can cause one component to have zero phase and the other component will have the difference in the two phases.

(a) $\mathbf{E} = (j\hat{\mathbf{x}} + \hat{\mathbf{y}})e^{-jkz}$

(b) $\mathbf{E} = [(1+j)\hat{\mathbf{y}} + (1-j)\hat{\mathbf{z}}]e^{-jkx}$

(c) $\mathbf{E} = [(2+j)\hat{\mathbf{x}} + (3-j)\hat{\mathbf{z}}]e^{-jky}$

(d) $\mathbf{E} = (j\hat{\mathbf{x}} + j2\hat{\mathbf{y}})e^{-jkz}$

Solution

Problem 5

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